AISES Software Evaluation Report

Executive Summary

Metric	Value
Total Requirements	12
Implemented	11
Partially Implemented	0
Not Implemented	1
Overall Compliance	91.67 %

Requirement Evaluation

AC-001 : Support numeric input (0-9), decimal point and multi-digit numbers.

Digits 0 through 9 append correctly to the current input display.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• Buttons for digits 0 through 9 are correctly mapped to lambda functions calling the digit method.

• The digit method correctly handles new inputs and appending to existing display text.

• Initial '0' display or new input states are correctly overwritten or appended to.

Issues

None

Recommendations

None

AC-002 : Support numeric input (0-9), decimal point and multi-digit numbers.

Only one decimal point is allowed per operand.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The `decimal` method correctly checks if `'.'` is already present in the current display string before appending a new decimal point.

• Handles `new_input` states properly by initializing the display to '0.' when a decimal is pressed on a fresh operand.

Issues

None

Recommendations

None

AC-003 : Support numeric input (0-9), decimal point and multi-digit numbers.

Multi-digit numbers are supported and displayed correctly as sequential digits are entered.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The `digit` method correctly checks if new input is expected or if the current display is '0', updating the value or appending sequential digits properly.

• The `decimal` method ensures a single decimal point is allowed and correctly handles appending it to multi-digit numbers.

Issues

None

Recommendations

None

AC-004 : Perform Addition (+), Subtraction (-), Multiplication (*) and Division (/).

Addition operation calculates the sum of two or more operands correctly.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The addition operator is properly bound in the UI button data list.

• The calculate method uses a dictionary mapping containing the '+' operation correctly using self.first + second.

• Calculated sums correctly update the display and support chaining operations via self.first = result.

Issues

None

Recommendations

None

AC-005 : Perform Addition (+), Subtraction (-), Multiplication (*) and Division (/).

Subtraction operation calculates the difference between operands correctly.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The subtraction operator is mapped correctly in the button data.

• The calculate method properly handles the subtraction operation using `self.first - second`.

Issues

None

Recommendations

None

AC-006 : Perform Addition (+), Subtraction (-), Multiplication (*) and Division (/).

Multiplication operation calculates the product of operands correctly.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The multiplication operator is correctly mapped in the operations dictionary as `'*': lambda: self.first * second`.

• Operands are correctly converted to float types prior to calculation to handle decimals.

Issues

None

Recommendations

None

AC-007 : Perform Addition (+), Subtraction (-), Multiplication (*) and Division (/).

Division operation calculates the quotient of operands correctly.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The division operator '/' is correctly mapped in operations dictionary to perform self.first / second.

• Exception handling is present to catch runtime errors such as division by zero.

Issues

None

Recommendations

None

AC-008 : Provide Clear (C) to reset all calculator state.

Clear (C) button resets all calculator state, memory, and display to initial zero or empty state.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The clear method correctly resets display to '0'

• Stored operand self.first is reset to 0.0

• Operator self.operator is set to None

• Input state self.new_input is reset to True

Issues

None

Recommendations

None

AC-009 : Provide Delete (DEL) to remove the last entered character.

Delete (DEL) button removes the last entered character from the input display.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The 'DEL' button is correctly mapped to the delete method.

• The delete method properly slices the string to remove the last character when length > 1.

• Edge case handled where single character input resets the display to '0'.

Issues

None

Recommendations

None

AC-010 : Validate division by zero BEFORE division and display an error message instead of performing the calculation.

Division by zero is detected before calculation and triggers an error message display instead of executing the operation.

Status : Not Implemented

Coverage : 0 %

Confidence : 100

Matched Files

None

Strengths

None

Issues

• No source files were mapped to this acceptance criterion.

Recommendations

• Provide or correctly map the relevant source implementation.

AC-011 : Evaluate the expression when '=' is pressed.

Equals (=) button triggers the evaluation of the current mathematical expression.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• The '=' button is successfully mapped to the 'calculate' method in the button configuration list.

• The 'calculate' method evaluates the expression using stored operands and operators.

• Handles errors gracefully during evaluation.

Issues

None

Recommendations

None

AC-012 : Display formatted numeric results.

Calculated results are formatted correctly and presented in the display field.

Status : Implemented

Coverage : 100 %

Confidence : 100

Matched Files

• Calculator.py

Strengths

• Results are correctly formatted based on whether they are integers or floating-point numbers.

• The formatted result is properly assigned to the display variable using self.display.set().

Issues

None

Recommendations

None